

TALAT KARASAKAL

Computer Engineering Student

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PROFILE

I find explaining why a system was built the way it was harder — and more valuable — than getting it to run. I study computer engineering and work mostly on desktop applications and image processing. The part of a task that interests me most is where the options get compared: which model, which library, which data setup was chosen and why. Longer term, I aim to work in the defence industry and enterprise technology.

TECHNICAL SKILLS

Languages: Java, Python, C, C++, C#

Frameworks and Libraries: Flask, JavaFX, Ultralytics YOLOv8, OpenCV, PySide6, React

Databases: SQLite, SQLAlchemy

Tools: Git / GitHub, Gradle, Maven, Docker, Mermaid, Unity

Testing: JUnit 5, JaCoCo

Domains: Desktop and web application development, computer vision and object detection, data modelling, performance and persistence design

Process: Requirements engineering (IEEE 830 / IEEE 1016), project management and planning, event organisation and coordination

EDUCATION

Izmir University of Economics — BSc in Computer Engineering

Izmir, Turkey | 09/2022 – 06/2027 (expected) | English-medium instruction | 50% merit scholarship

EXPERIENCE

Probel Yazılım — Computer Engineering Intern

Izmir, Turkey | 08/2026 – 09/2026 | IT and R&D Department

- At a healthcare information systems company, ran a web-based survey application end to end as a standalone R&D project, independent of the company codebase, during the mandatory summer internship; selected the process model and technology stack.
- Prepared a 49-page requirements and design document following IEEE 830 and IEEE 1016: 66 functional requirements (77 including sub-items), 20 use cases and a traceability matrix.
- Selected Flask, SQLAlchemy and SQLite, ruling out Django with a documented rationale; defined a nine-entity data model and a co-ownership model with five separate permissions.
- Identified that client-side validation could be bypassed and decided to enforce all checks on the server; identified a privilege-escalation chain arising from admin permissions and restricted admin grants to the primary owner.

Izmir University of Economics — Student Assistant, Directorate of Corporate Communications

Izmir, Turkey | 08/2025 – 12/2025, 02/2026 – 07/2026

- Supported on-site coordination of the university's large-scale academic and corporate events within the Events Unit: the academic year opening, commencement ceremonies, the Long Service Awards and the 25th anniversary celebrations.

SCA Social — Project Management Intern (Volunteer)

Istanbul, Turkey (Remote) | 01/2026 – 02/2026

- Prepared annual budget planning and cost tables for a defence-industry context; designed an AI-based simulation approach for early identification of risks in construction projects.
- Turned data protection (KDPL) processes into formal documentation through case analysis; produced a Project Initiation Document and Gantt chart covering a project's scope, resource and time planning.

Izmir University of Economics — Admissions and Outreach Assistant

Izmir, Turkey | 07/2024 – 09/2024, 07/2025 – 08/2025

- Advised prospective students and parents in the call-centre and transfer units, and ran campus tours.

PROJECTS

Nexa — Smart Road Safety with 5G and AI

TEKNOFEST 2026 (Turkcell) | 3-person team, AI owner | 02/2026 – 07/2026

- Made and validated the technical decisions for an inference pipeline detecting vehicle type, licence-plate region, colour, driver violations and seat occupancy from a fixed roadside camera using a single 21-class YOLOv8s model.

- Remapped 10 open-source datasets onto the competition's 21 official classes and annotated team-recorded footage for classes with no open-source equivalent, assembling a 32,425-image dataset while preventing leakage from consecutive frames at the split stage.
- Benchmarked YOLOv8s against YOLOv8m on two GPUs and selected YOLOv8s: 0.701 overall mAP@0.5, 0.896 for plate-region detection, ~12 ms per frame. Cleared the Preliminary Design Report stage with 85.67/100.

Exam Scheduling and Management System

5-person course project (SE 302) | Java, JavaFX, SQLite, Gradle | 10/2025 – 12/2025

- Owned the interface and input-handling responsibility in a desktop application automating exam-week scheduling, and designed the entire interface. Recommended Java to the team; accepted.
- Proposed parallelising the scheduling computation, which substantially reduced runtime. Proposed a persistence approach storing recently opened files and filters, removing the need to recompute on every launch.

Integrated Assignment Environment

5-person course project (CE 316) | Java, JavaFX, SQLite | 05/2026 – 06/2026

- Selected the execution-engine module within a five-module team split, for an application that compiles and runs student submissions in an isolated process and compares output against expected results; process management, timeout control, runtime measurement, compiler path validation and error classification fell within this module.
- Three unit-test files were added for the module; one caught a defect where submissions failing at runtime were counted as passing.

Discrete Event Simulation

4-person course project (SE 116) | Java, JUnit 5, Maven | 04/2024 – 05/2024

- Worked on the input-parsing layer of a Java application that parses and validates two configuration files and reports station queues and delay metrics; identified and fixed parsing-loop defects in the JobFileParser and WorkflowFileParser classes.
- After course submission, a personal layer was added to the repository: a Maven migration, JUnit 5 test infrastructure and a Swing interface. The simulation core produced by the team's work was preserved.

VOLUNTEER EXPERIENCE

IEU Software Community — Member, Audit Board Member

Membership 10/2024 – Present | Audit Board 09/2025 – 07/2026

Industrial Systems Society — Member, event organisation team

10/2024 – Present

CERTIFICATIONS AND TRAINING

AI and Data: AI Fluency for Students, Claude Platform 101, Introduction to Model Context Protocol, Model Context Protocol: Advanced Topics (Anthropic, 2026) · Introduction to Modern AI, Introduction to Data Science (Cisco, 2024) · Introduction to Machine Learning Workshop (IUE, 2025)

Project Management: Agile Project Management, Project Management with Microsoft Project (BTK Academy, 2026)

Quality and Information Security: ISO 9001:2015 Quality Management Systems (Sigmacert Global, 2026) · Fundamentals of Information Security (IUE, 2024)

Industry and Tools: Career and Competency Meetings, Defence Industry Academy (2024–2026) · Kuika Designer and Builder Course (2025) · App Designer Workshop (Figes, 2025)

LANGUAGES

Turkish — Native · English — B2 · German — A2